

Image Quality Examiner Document

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Purpose

Image Quality Examiner (IQE) is a program developed by the 2026 WPI Team for The Postal Museum (TPM) in London. It aims to assist with quality assessment of TPM's images in the PhotoShelter library. This document outlines the technical details of IQE and instructions on using the program.

Main Functionality

IQE can search through any gallery and collection in the PhotoShelter library to find images that need to be reviewed based on its metadata. Then it can tag images by attaching keywords to those images and showing reasons for review. Those keywords are very searchable in PhotoShelter, and they can be easily removed after the image has been reviewed.

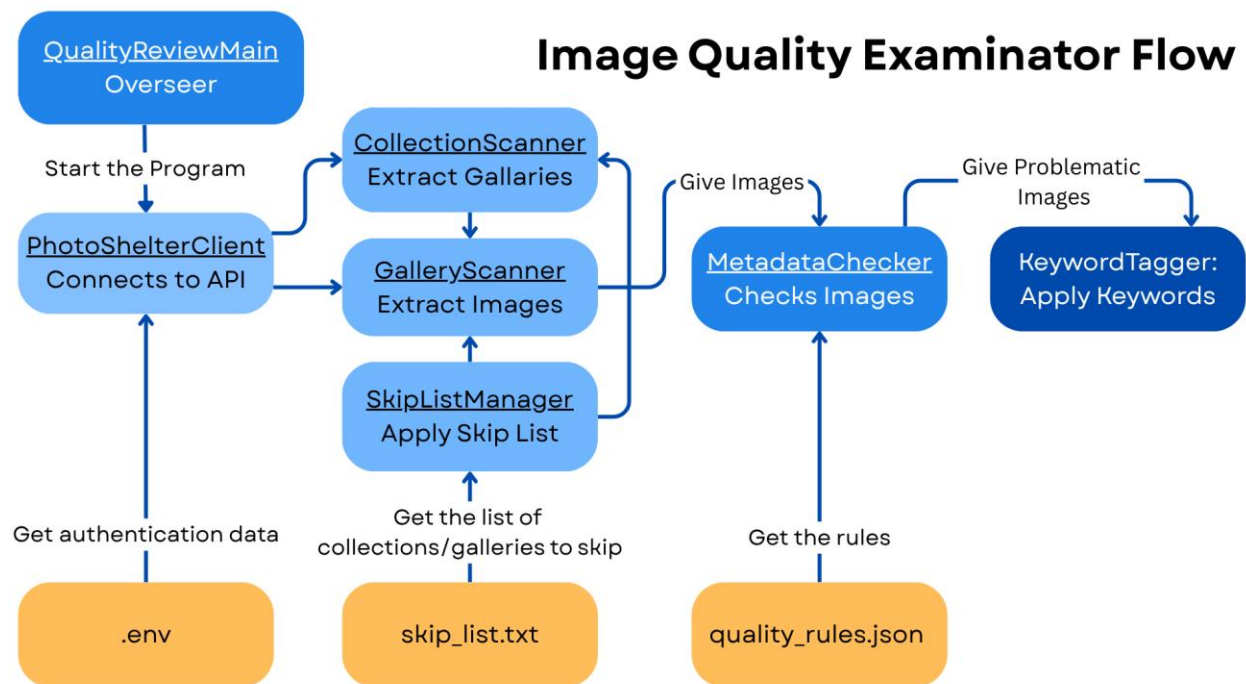
Specification

Language	Python 3.14
Integrated Development Environment (IDE)	PyCharm 2025.3.3
Files	QualityReviewMain.py PhotoShelterClient.py CollectionScanner.py GalleryScanner.py MetadataChecker.py quality_rules.json KeywordTagger.py .env skip_list.txt SkipListManager.py

File Review

File Name	Function
QualityReviewMain.py	This is the main orchestrator of IQE. It brings every component together and executes the program. The file hosts the user interface for using the program.
PhotoShelterClient.py	It handles connections and communication with the PhotoShelter API. It authenticates the program, sends requests, and manages the rate limit. Authentication setup in the .env file is required.
CollectionScanner.py	It extracts all galleries from a given collection, then use the gallery scanner to extract all the images. It also includes functions that extract galleries from a collection. No setup is needed.
GalleryScanner.py	It extracts all images from a given gallery. No setup is needed.
MetadataChecker.py	It receives a list of images from the scanners. It checks the images for quality violations and flags them. It loads the metadata checks from quality_rules.json. Ensure quality_rules.json is loaded. To add new rules, modification to the function _apply_rule is needed.
quality_rules.json	It contains all of the metadata rule checks. Modify the “enabled” value to enable/disable specific rules. Reference the existing rule structure when adding new rules.
KeywordTagger.py	From MetadataChecker, it receives a list of images along with their violations. Based on the list, it uploads the violation keywords of each image to the PhotoShelter library.
.env	It stores sensitive authentication information. If the .env file does not exist, create one with the following template: Template: <pre>API_KEY=[TPM'S API KEY] EMAIL=[TPM ACCOUNT] PASSWORD=[PASSWORD OF THE TPM ACCOUNT] ORG_ID=[TPM'S ORGANIZATION ID ON PHOTOSHELTER]</pre> Example: <pre>API_KEY=12FyU EMAIL=tpm@gmail.com PASSWORD=password ORG_ID=uIj62</pre> If you do not know the API key, ask the WPI team or Louis.

skip_list.txt	This is a list of collections and galleries to skip when doing image quality assessment. Users may add and remove names of the collections and galleries for their images to not be checked. Users may choose to apply or ignore the skip list in QualityReviewMain.
SkipListManager.py	It manages the skip list. It loads the list from the file and tells the scanners whether to skip this collection or gallery.



How to run the IQE?

1. Download the files above. Ask Louis on how to get them.
2. Install Python 3.14 on the computer. The installation link is <https://www.python.org/downloads/release/python-3145/>
3. Recommend using an IDE to run the program. This program was developed on PyCharm: <https://www.jetbrains.com/pycharm/>. Otherwise, run it in an IDE of your choice. Visual Studio Code was confirmed to be a working platform.
4. Check for issues while opening the folder in the IDE. Common tasks include installing necessary packages, setting up a Python 3.14 interpreter, and creating the `.env` file.
5. Also recommend looking around and seeing the comments on the files.
6. To execute the program, go to `QualityReviewMain.py` and scroll to the bottom for the main function. There, you can configure the run setting like collection/gallery name, mode, dry run, and apply skip list.

7. Once you configure the run, execute `QualityReviewMain.py`. While it is running, check the console messages. They are progress updates and can tell you what's wrong.

Examine Images on PhotoShelter

When a run is complete and keywords are uploaded, you can see the effect immediately. Go on PhotoShelter library to examine the images. Some things to consider:

1. Every tagged image will have the keyword "quality-review-needed". On top of that, each image will also have specific keywords like "low-resolution" or "auto-exposure" to show what metadata rule it failed.
2. You may search the keywords directly to see all the images that have them. For example, search "quality-review-needed" to see every image that needs to be reviewed.
3. Keywords can be easily removed by editing metadata on the website.